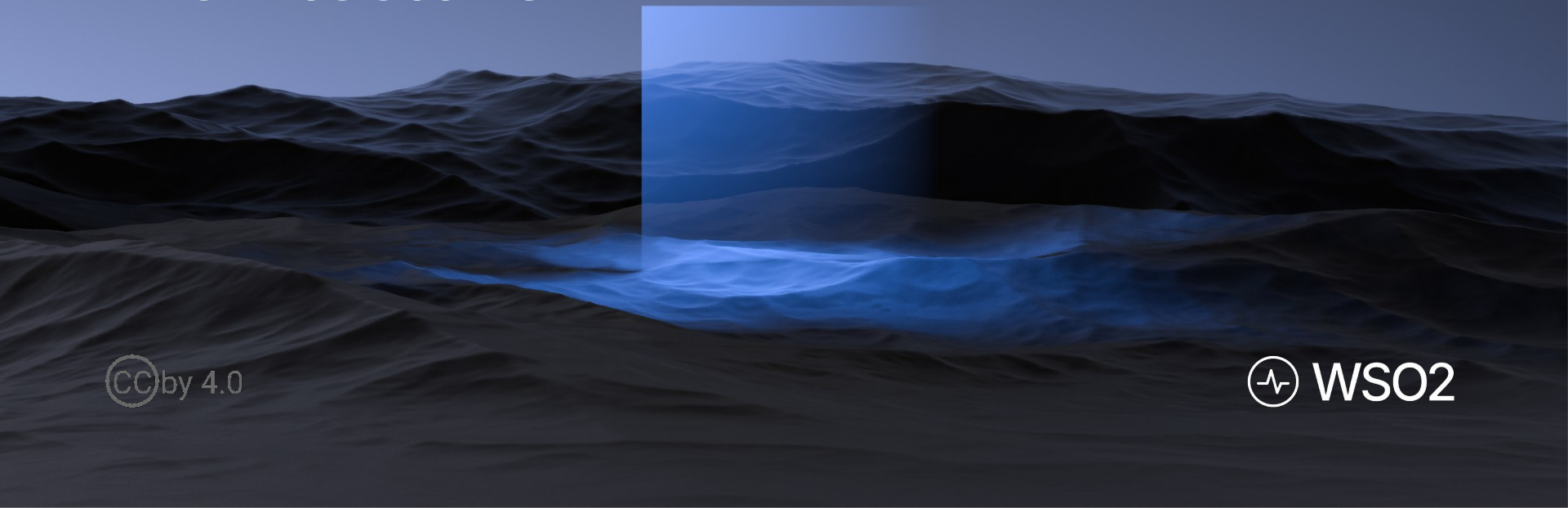
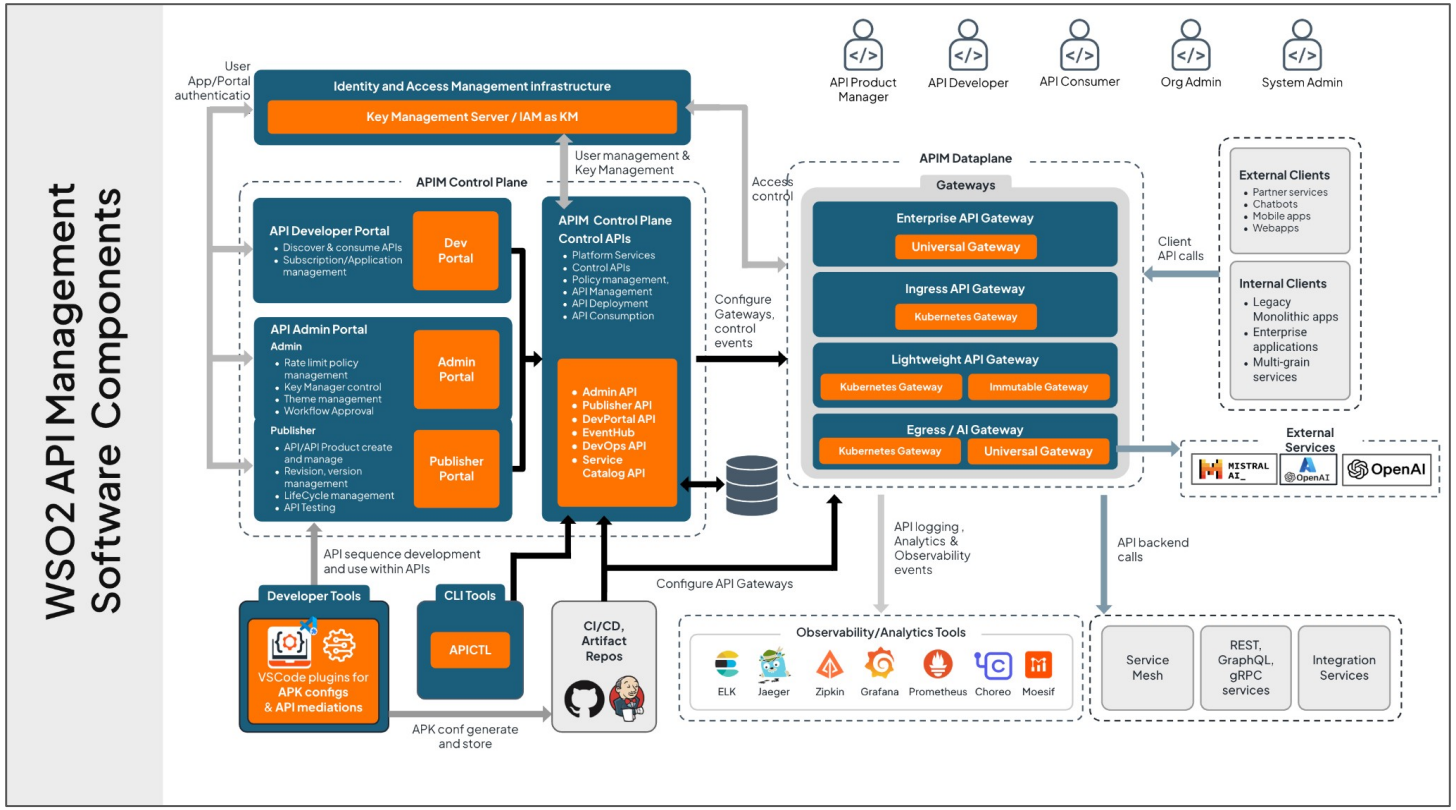
Architecture



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WSO2

Components


WSO2 API Management Software Components
User App/Portal authenticatio
Identity and Access Management infrastructure
Key Management Server / IAM as KM
User management & Key Management
API Product Manager
API Developer
API Consumer
Org Admin
System Admin
APIM Control Plane
API Developer Portal
• Discover & consume APIs
• Subscription/Application management
Dev Portal
API Admin Portal
Admin
• Rate limit policy management
• Key Manager control
• Theme management
• Workflow Approval
Admin Portal
Publisher
• API/API Product create and manage
• Revision, version management
• LifeCycle management
• API Testing
Publisher Portal
APIM Control Plane Control APIs
• Platform Services
• Control APIs
• Policy management,
• API Management
• API Deployment
• API Consumption
• Admin API
• Publisher API
• DevPortal API
• EventHub
• DevOps API
• Service Catalog API
Configure Gateways, control events
Access control
APIM Dataplane
Gateways
Enterprise API Gateway
Universal Gateway
Ingress API Gateway
Kubernetes Gateway
Lightweight API Gateway
Kubernetes Gateway
Immutable Gateway
Egress / AI Gateway
Kubernetes Gateway
Universal Gateway
Client API calls
External Clients
• Partner services
• Chatbots
• Mobile apps
• Webapps
Internal Clients
• Legacy Monolithic apps
• Enterprise applications
• Multi-grain services
External Services
MISTRAL AI_
OpenAI
OpenAI
API sequence development and use within APIs
Developer Tools
VSCode plugins for APK configs & API mediations
CLI Tools
APICTL
CI/CD, Artifact Repos
APK conf generate and store
Configure API Gateways
API logging , Analytics & Observability events
API backend calls
Observability/Analytics Tools
ELK
Jaeger
Zipkin
Grafana
Prometheus
Choreo
Moesif
Service Mesh
REST, GraphQL, gRPC services
Integration Services

API Control Plane

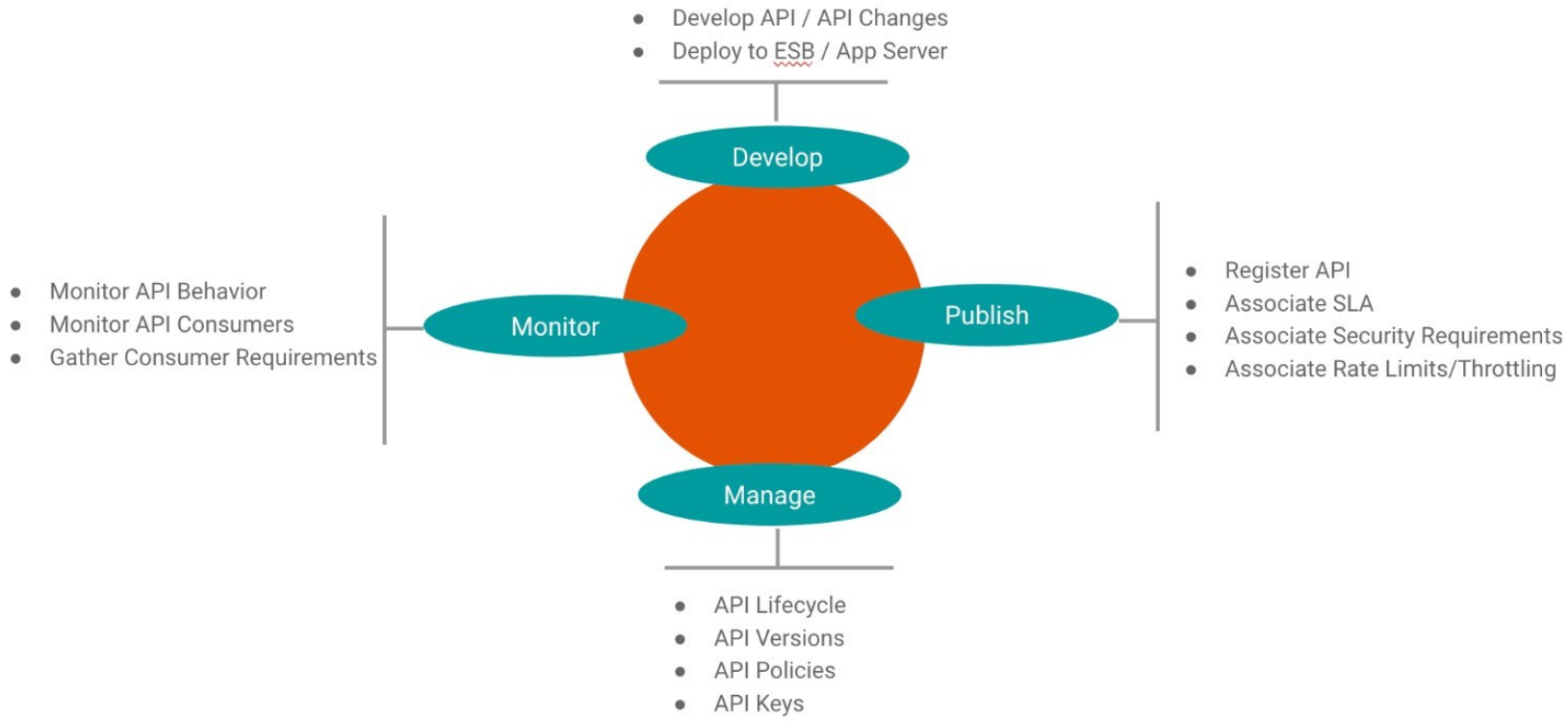
WSO2

API Control Plane

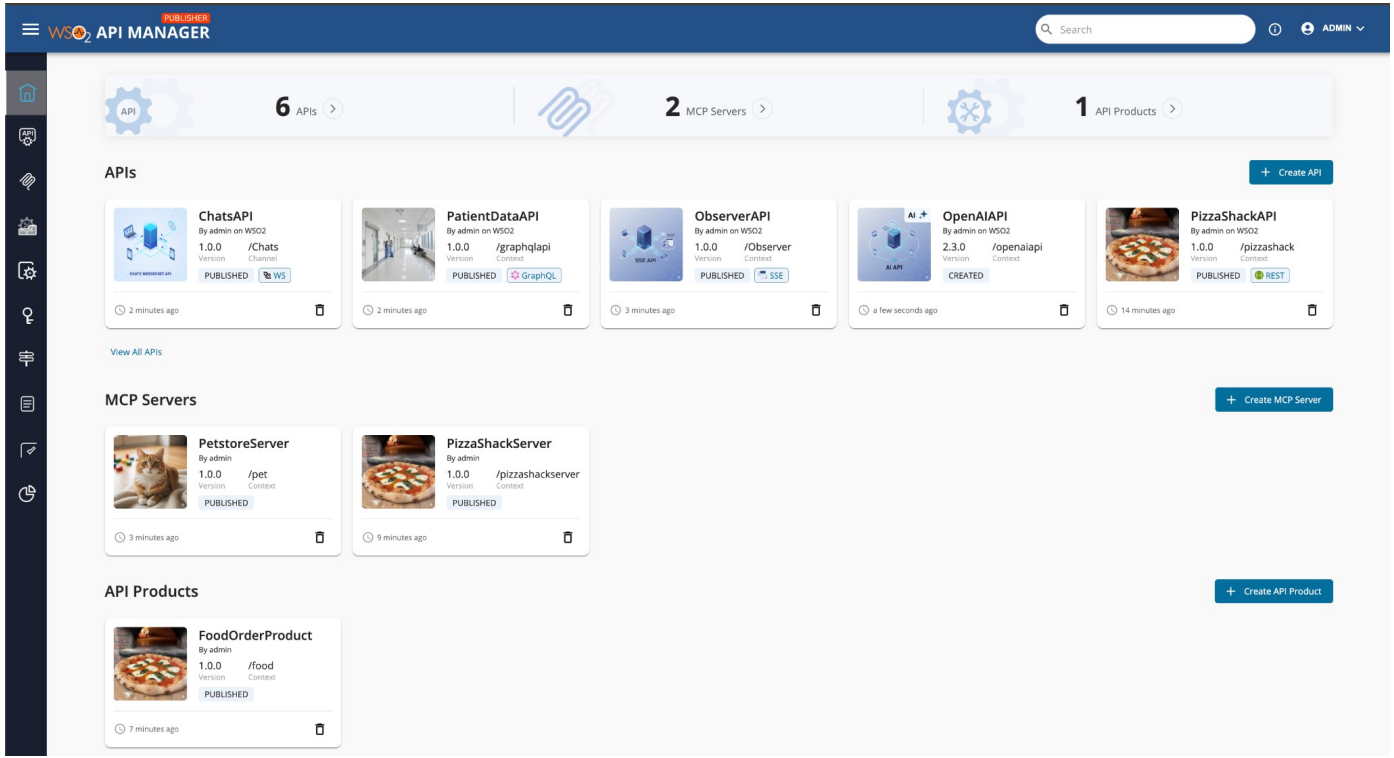
API Publisher Portal

WSO2

API Publisher Portal



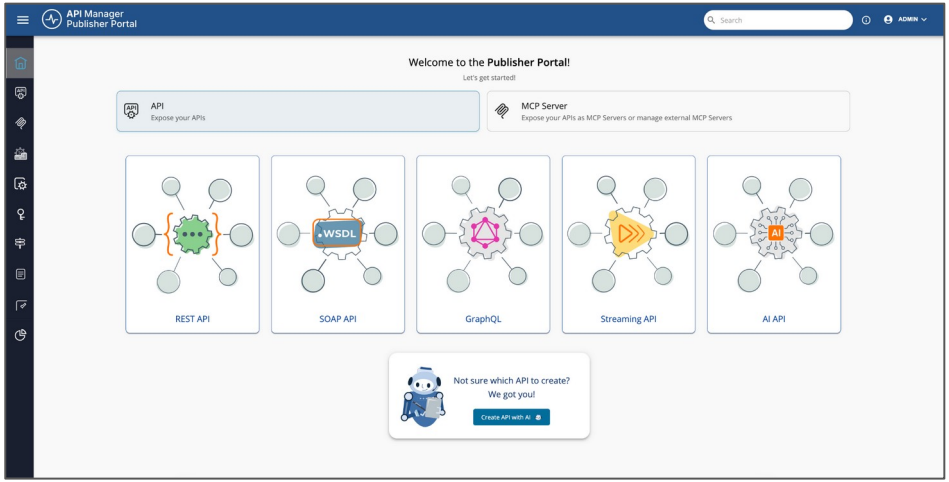
API Publisher

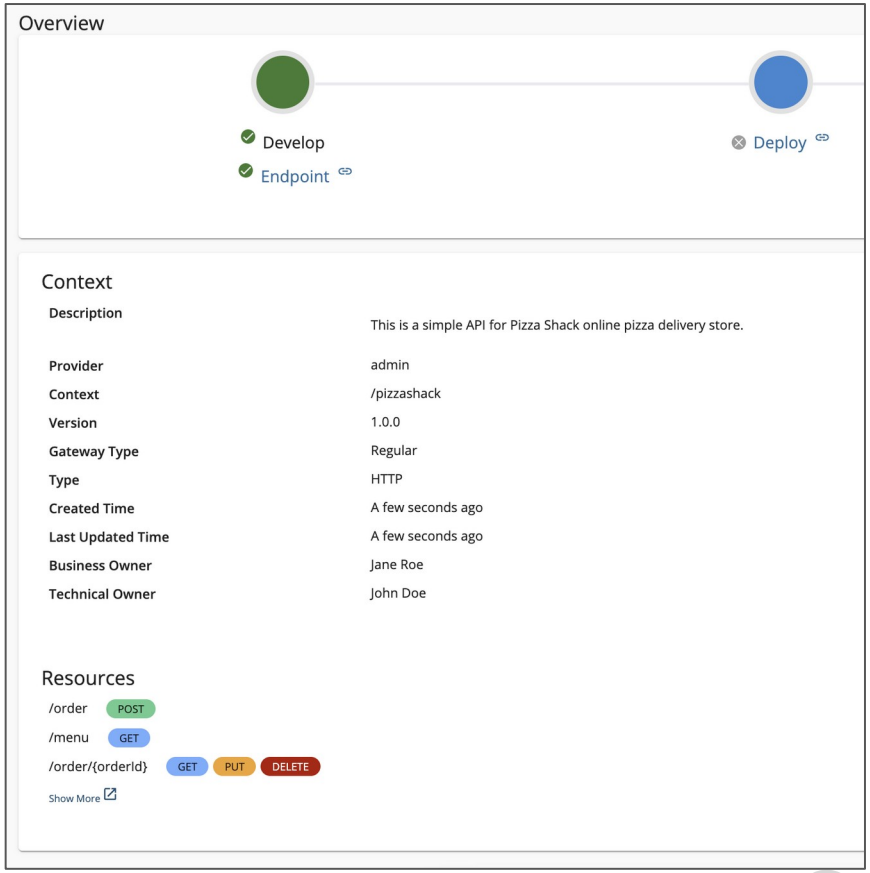


Link - [API Publisher]

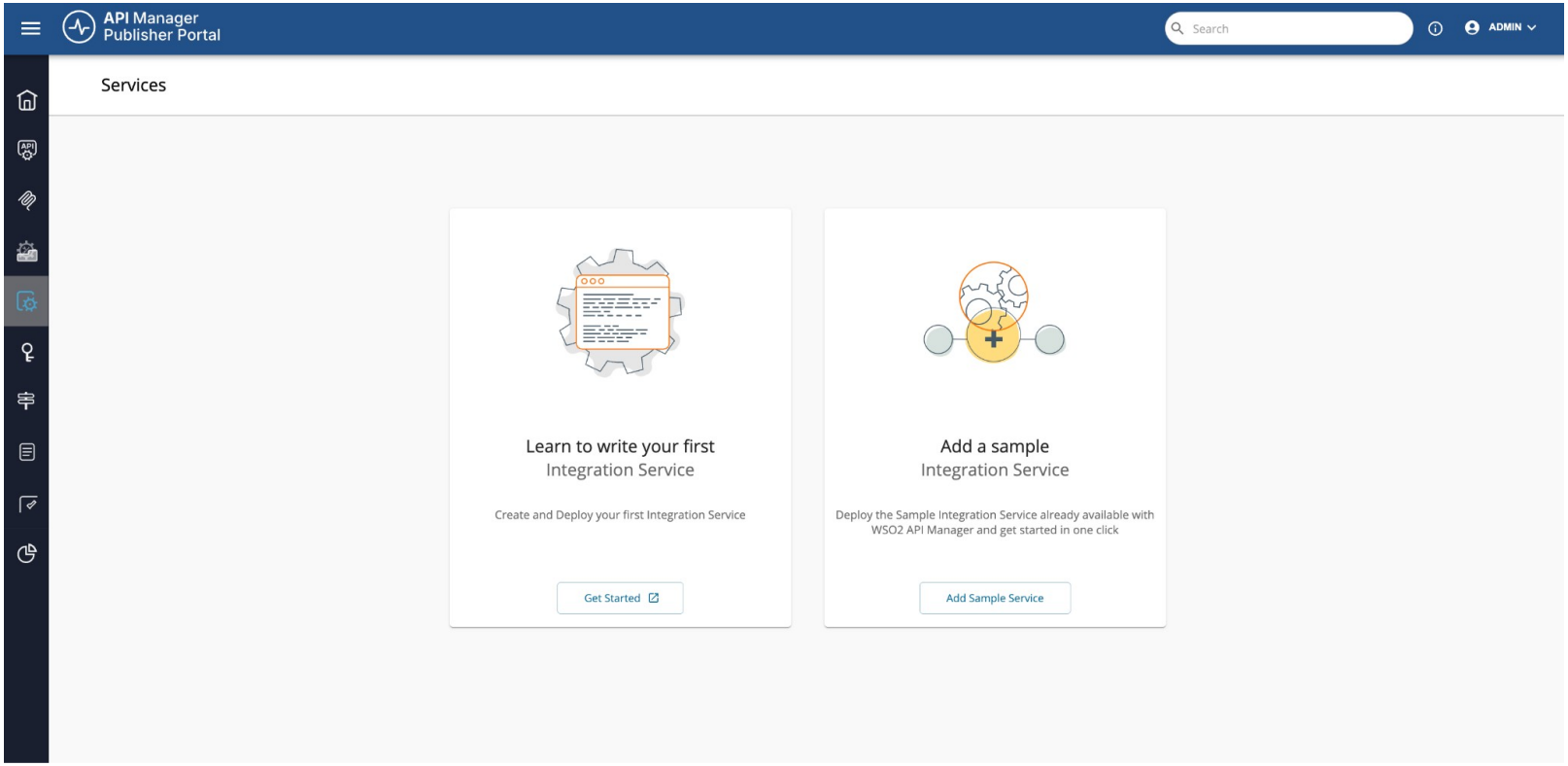
Usage of API Publisher







Service Catalog

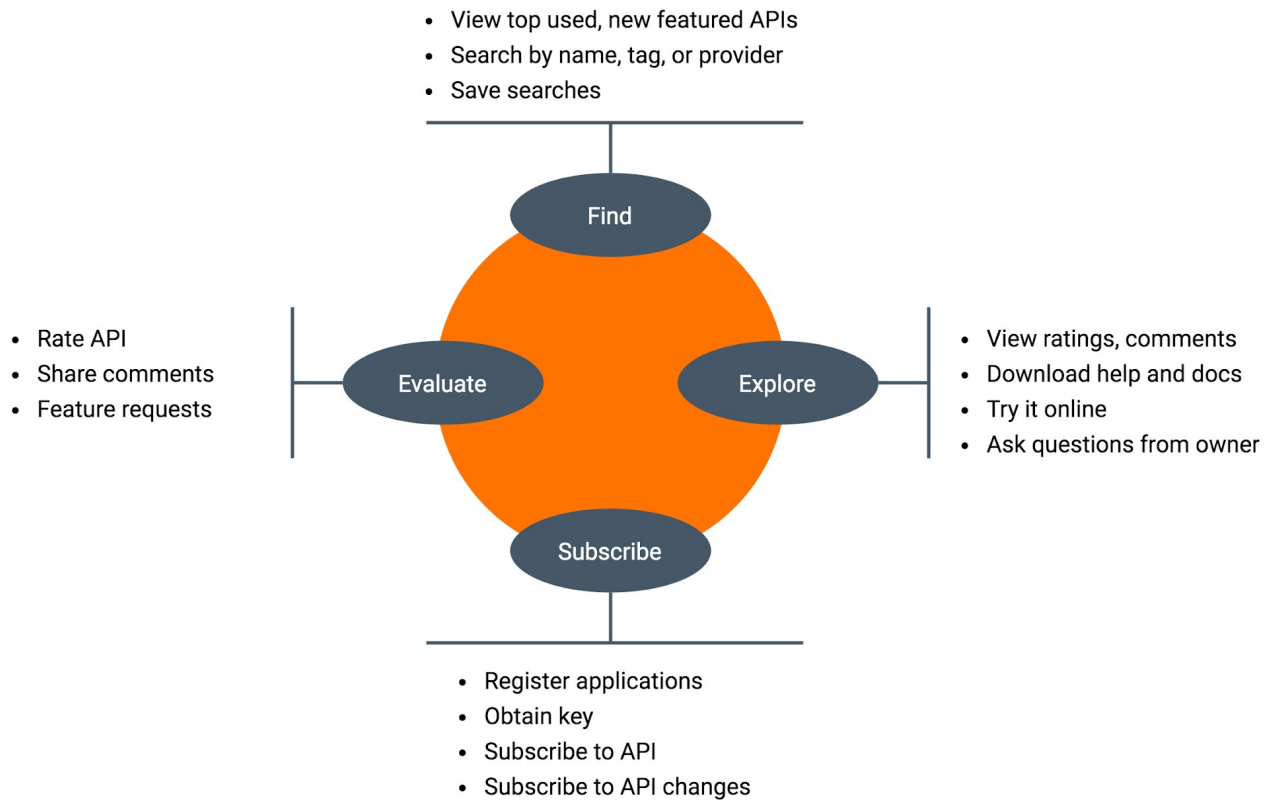


API Control Plane

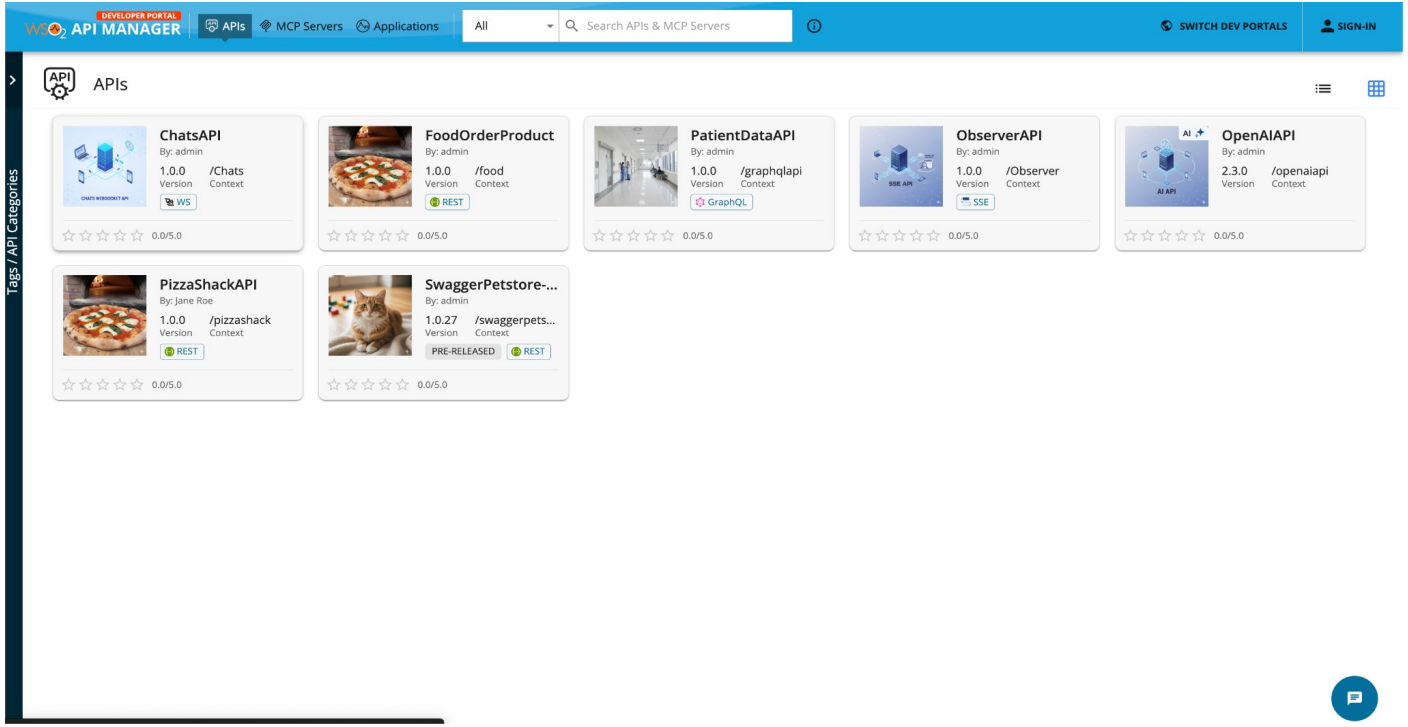
API Developer Portal

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API Developer Portal

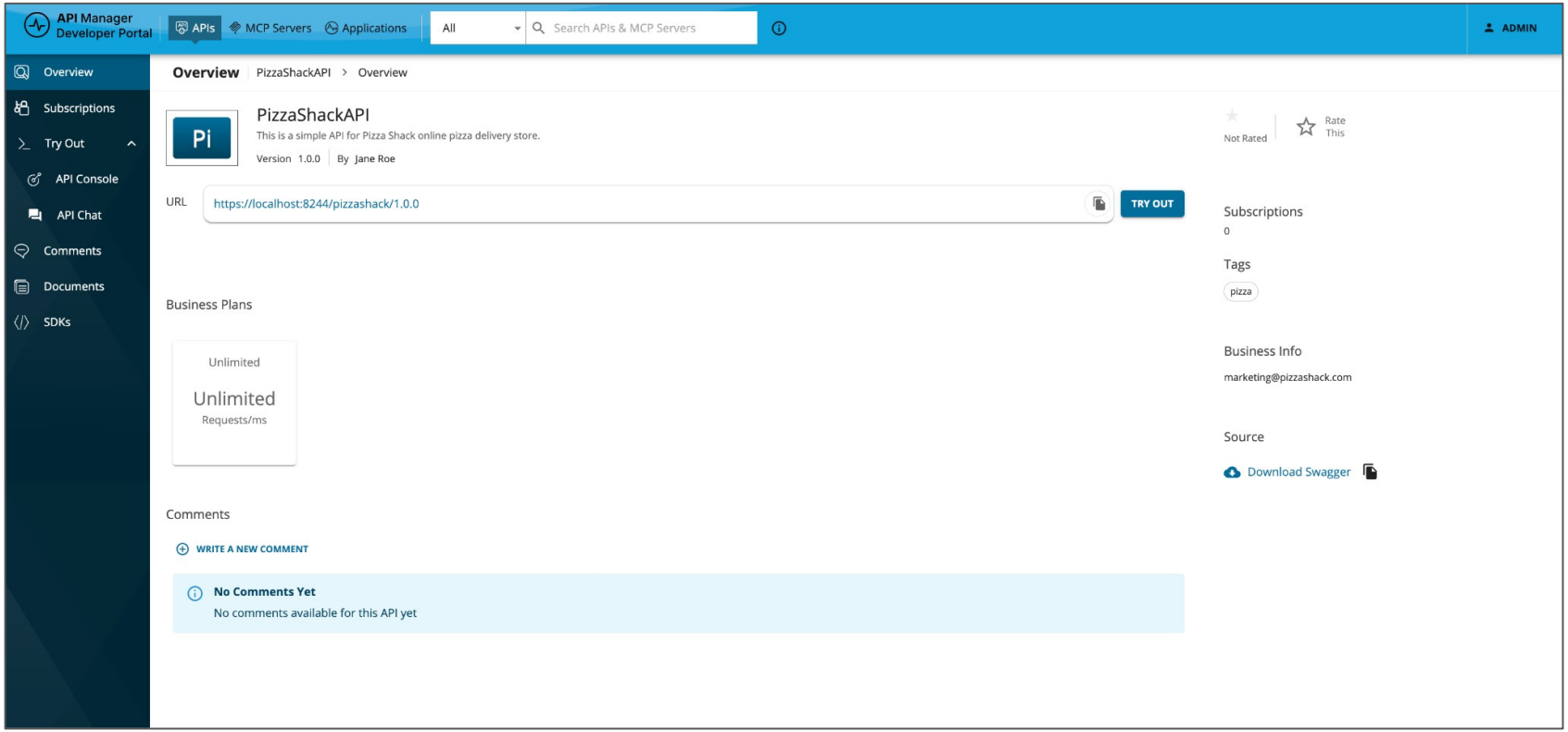


API Developer Portal

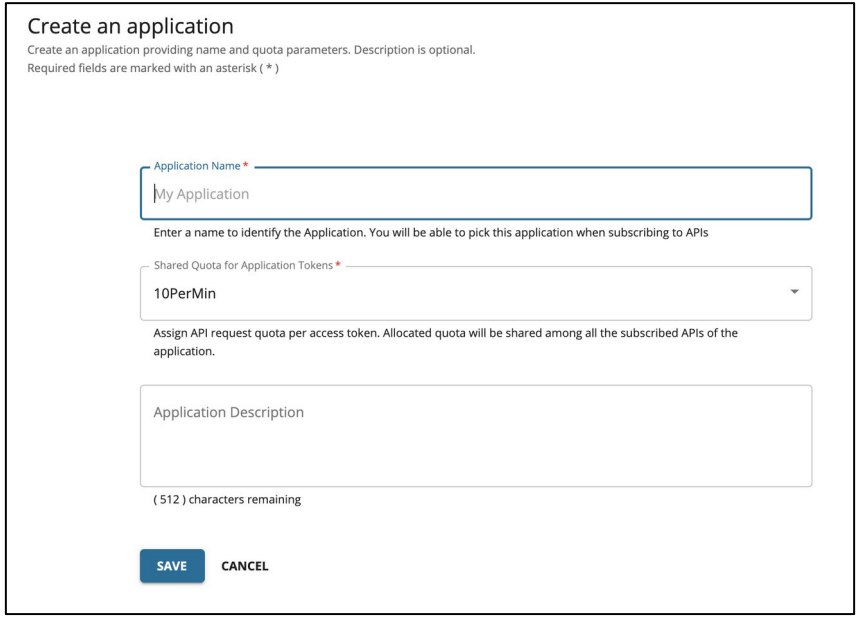


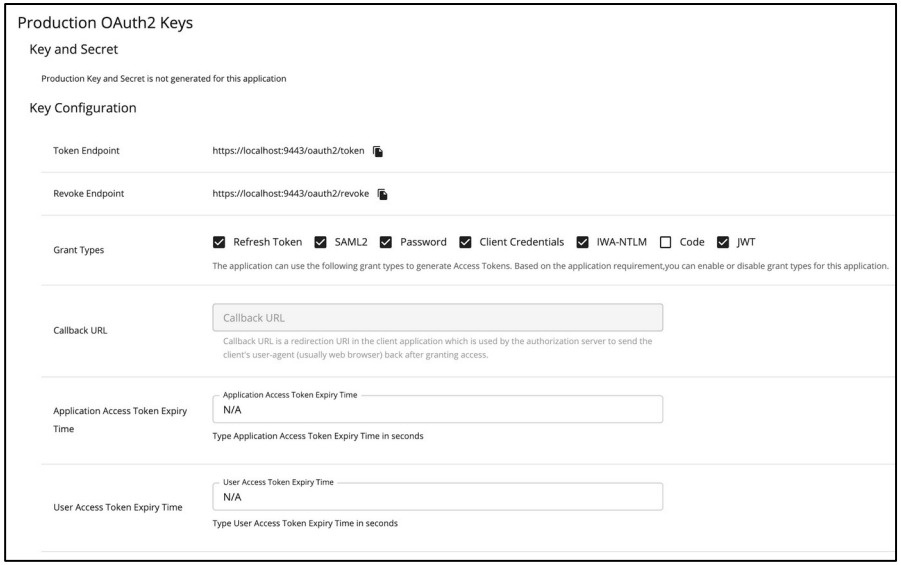
Link - [API Devportal]

Usage of API Developer Portal



Usage of API Developer Portal



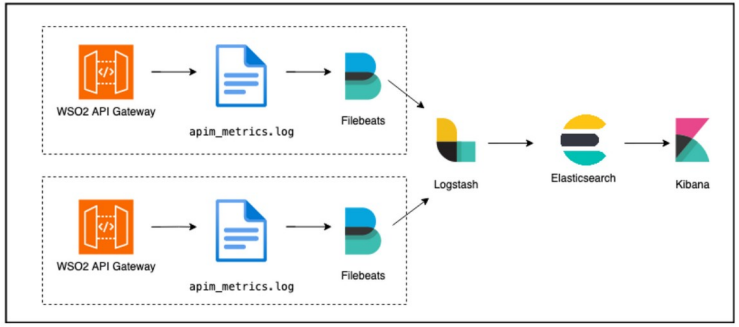


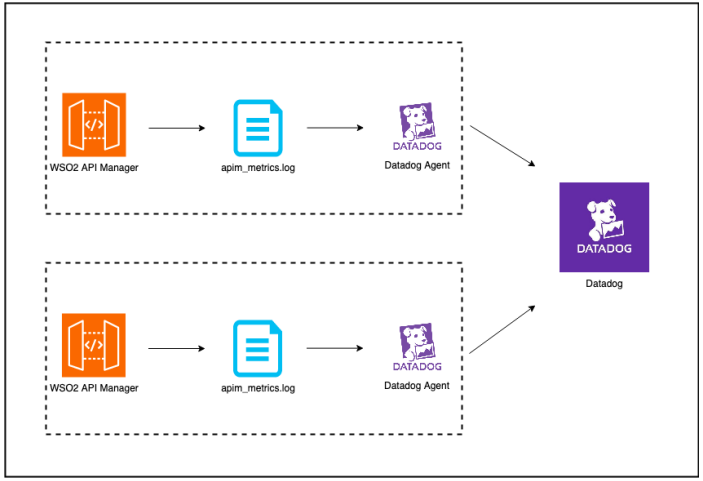
Analytics

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Analytics

- Analyses API requests for statistical reporting and abnormality detection
- Moesif Analytics. Refer Moesif Analytics Integration.
- ELK and Datadog based analytics for on-prem analytics. Refer ELK and Datadog guide.

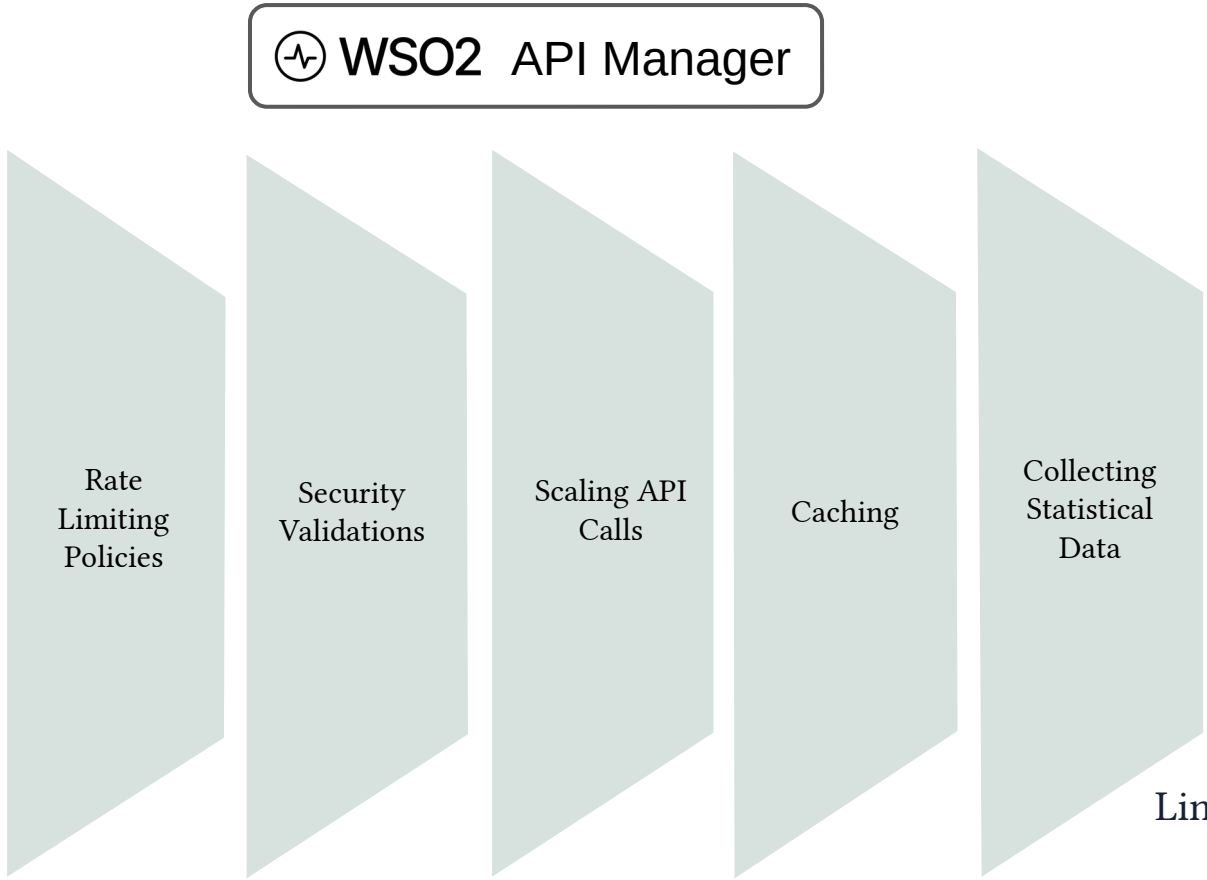




Classic Gateway

WSO2

Classic Gateway

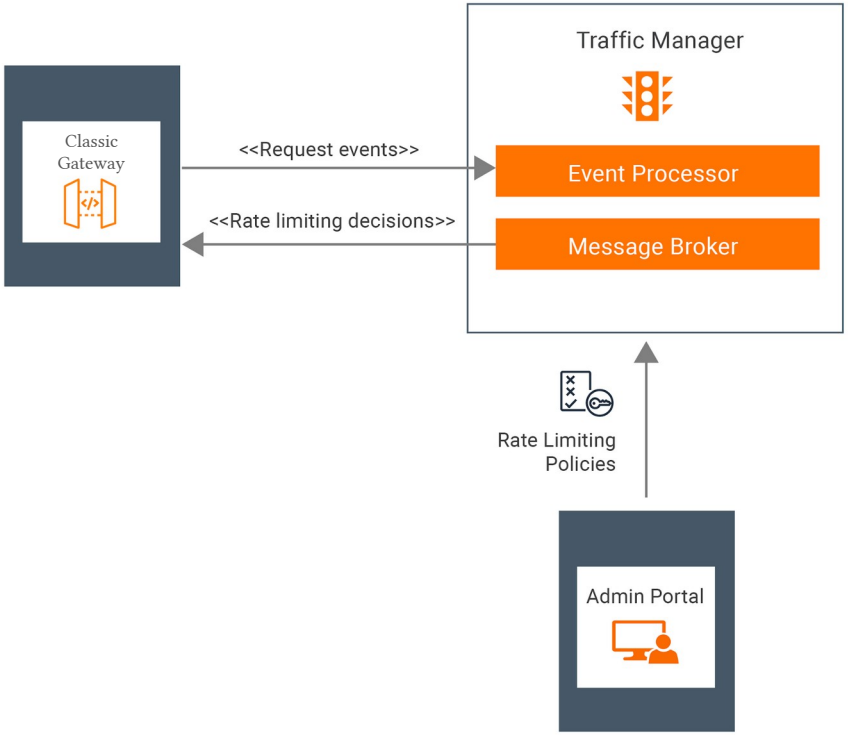


Link - API Gateway

Traffic Manager

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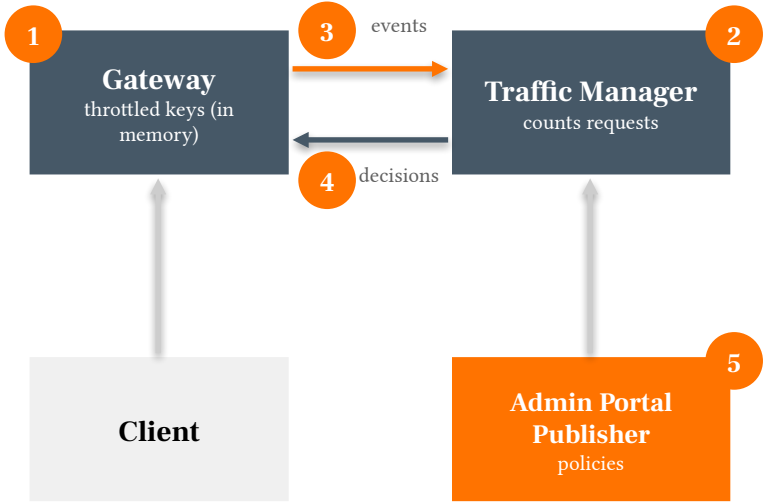
Traffic Manager



Keeps a count of API requests flowing through the API Gateways against various policies and ensures the rate limits (quotas) are met.

Link - Traffic Manager

Traffic Manager: Main Parts



Ports: 9611 (TCP) / 9711 (SSL) for events, 5672 for the throttleData JMS topic.

1. **Gateway.** Receives every API call. Its throttle check reads a local list of throttled keys, which is kept in memory.

2. **Traffic Manager.** Counts requests and decides when a limit is crossed. In an all-in-one server it runs inside the same APIM; in a distributed setup it is a separate profile.

3. **Event stream, gateway to TM.** Each gateway sends one small event per request to the TM, on port 9611 (TCP) or 9711 (SSL).

4. **JMS topic, TM to gateway.** The TM publishes its decisions on a topic called throttleData, on port 5672. Every gateway listens to that topic.

5. **Admin Portal and Publisher.** This is where you define the policies: Advanced, Subscription, Application, Custom and Deny. When you save a policy, it is deployed to the TM as counting logic.

Let’s try it out!

Getting Started with WSO2 API Manager

WSO2